SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	S.Padmaja	**Reg. No.:**	192424159
Section / Batch:	AI&DS	**Date:**	16-07-2026

Code of Conduct

I ______S.Padmaja____________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	NLP Models, Language Processing Evolution	TP-01
Comparative Study of NLP Models and Algorithms	Model Analysis, Algorithm Comparison	TP-02
Regular Expressions for Real-World Text Processing	Pattern Matching, Information Extraction	TP-03
Designing Efficient Regular Expression Patterns for Information Extraction	Text Mining, Data Extraction Techniques	TP-04
Finite State Automata in Natural Language Processing	State Transitions, Language Recognition	TP-05
Building a Finite State Automaton for Token Recognition	Lexical Analysis, Token Validation	TP-06
Morphology and Word Formation in Natural Languages	Morpheme Analysis, Word Structure Processing	TP-07
Inflectional Morphology: Understanding Grammatical Variations	Grammatical Analysis, Word Normalization	TP-08
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Finite-State Morphological Parsing Techniques	Morphological Parsing, Finite-State Processing	TP-11
Morphological Analysis for Indian Languages	Language-Specific Morphology, NLP Challenges	TP-12
Porter Stemmer: Design, Working Mechanism, and Applications	Stemming Algorithms, Text Preprocessing	TP-13
Stemming vs Morphological Parsing: A Comparative Study	Word Reduction, Morphological Processing	TP-14
Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	End-to-End Language Processing Pipeline	TP-15

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %
14	30	19	20	15	10	94

Faculty In-charge	Course Coordinator	Program Director
Kumaragurubaran.T		
Signature: ________________	Signature: ________________	Signature: ________________
Date: 16.07.2026	Date: ____________________	Date: ____________________

DECODING LANGUAGE

How Machines Learned to Read, Listen & Talk Back

A journey through the models and algorithms of Natural Language Processing — from counting words to understanding meaning.

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Why Do Machines Need NLP?

Computers only understand numbers. Humans speak in words.

Humans Communicate In...

English	Tamil
Hindi	Telugu
French	**...and thousands more**

NLP Converts Language → Numbers

"I love Artificial Intelligence."

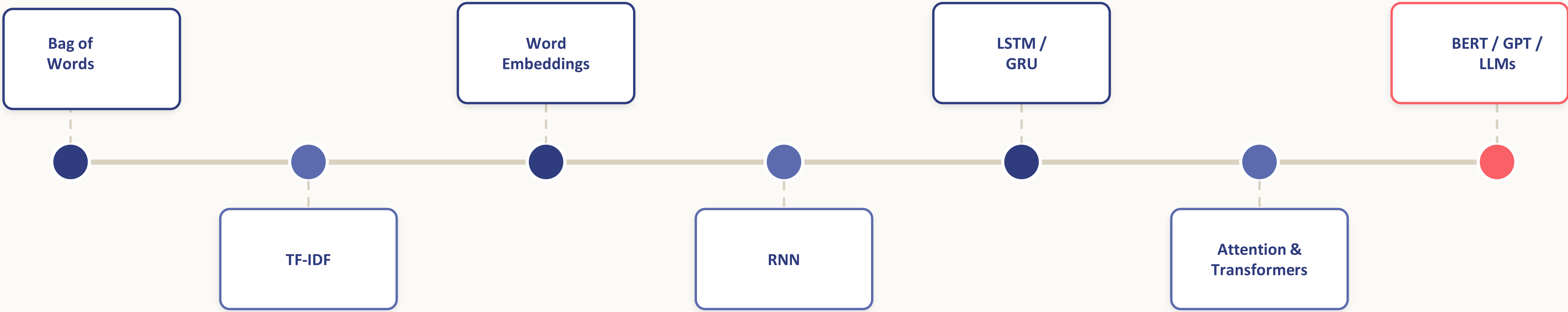
▼

`[0.42, -0.15, 0.81, 0.09, ...]`

This numeric form is what every model in this deck is built on top of.

The Evolution of NLP

Each stage solved the limitation of the one before it.



1. Bag of Words — Just Count the Words

#

Represents a sentence by counting how often each word appears. Grammar, order, and meaning are ignored.

Vocabulary: { I, love , to, read }

"I love to read "	→	`[1, 1, 1,1 ]`
“read to love I"	→	`[1, 1, 1, 1]`

Same meaning, identical vectors — BoW cannot tell the sentences apart.

✓ Advantages

- Very simple to implement
-
- Fast to compute
-
- Works well as a quick baseline

✗ Disadvantages

- Ignores word order & context
-
- Ignores grammar entirely
-
- Produces huge, sparse vectors

2. TF-IDF — Not All Words Matter Equally

Common words like "the" appear everywhere and carry little meaning. TF-IDF gives rare, informative words more weight.

TF-IDF = TF × IDF

`IDF = log( Total Documents ÷ Documents containing the word )`

D1: The movie is excellent
D2: The movie is good
D3: The movie is boring

Weight comparison across the 3 reviews:

"movie" — *appears in every document → very low IDF*

"the" — *appears everywhere → almost no weight*

"excellent" — *appears once, in D1 only → high IDF*

Used in: Search engines • Document ranking • Spam detection

3. N-Grams — Keeping a Little Order Back

BoW throws order away completely. N-grams group words into short sequences so some context survives.

Sentence: "I love NLP"

Unigrams

- `I`
- `love`
- `NLP`

Bigrams

- `I love`
- `love NLP`

Trigrams

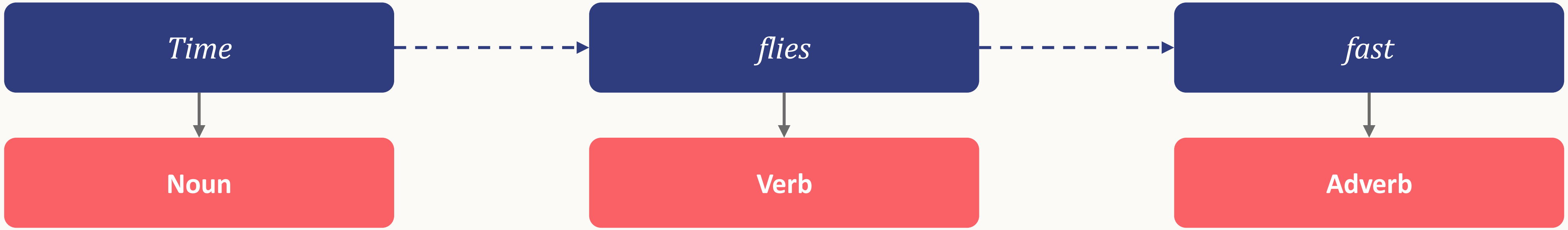
- `I love NLP`

Advantage: preserves short-range context. **Disadvantage:** vocabulary size explodes quickly.

4. Hidden Markov Models — Guessing the Grammar

Language is sequential — order changes meaning ("Dog bites man" ≠ "Man bites dog"). HMMs use this order for tasks like part-of-speech tagging.

Example — tagging "Time flies fast"



Transition Probability

The likelihood of moving from one tag to the next — e.g. how often a Noun is followed by a Verb.

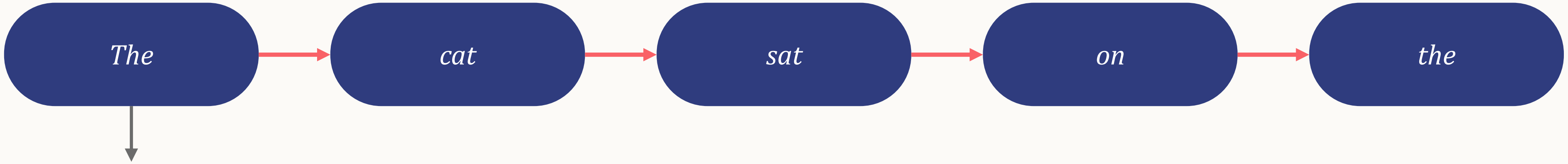
Emission Probability

The likelihood of a specific word appearing given a tag — e.g. how likely "flies" is, given the tag Verb.

The Viterbi Algorithm combines both to find the single most probable tag sequence.

5. RNNs — Machines That Remember

A Recurrent Neural Network reads text one word at a time, carrying a "memory" of everything seen so far.



predicts "mat" — using memory built up from all prior words

⚠ The Long-Sentence Problem

"The movie that I watched last week with my friends was absolutely fantastic."

"movie" and **"fantastic"** are far apart — by the time the RNN reaches the end, earlier memory has faded. This is the **Vanishing Gradient Problem**: gradients shrink during training, so the network struggles to learn long-range dependencies.

6. LSTM — A Smarter Notebook

Long Short-Term Memory networks fix the vanishing gradient problem using three gates that control what to remember.

1 **Forget Gate**

Erases information that is no longer useful.

2 **Input Gate**

Writes new, important information into memory.

3 **Output Gate**

Decides what part of memory to reveal now.

Example: Coreference across a long sentence

"The trophy didn't fit into the suitcase because it was too big."

An LSTM correctly keeps track that **"it"** refers to the **trophy** — not the suitcase — something a plain RNN often loses across the gap.

Putting It All Together

Model	Captures	Best Known For	Key Limitation
Bag of Words	Word frequency	Fast baselines	No order or meaning
TF-IDF	Word importance	Search & ranking	Still no context
Word2Vec / GloVe	Word meaning	Semantic similarity	One vector per word
HMM	Tag sequences	POS tagging	Short-range only
RNN	Sequential memory	Simple text prediction	Forgets long sentences
LSTM	Long-term memory	Long text understanding	Slow, still sequential
Transformers (BERT/GPT)	Full-sentence context, in parallel	Modern LLMs & chatbots	Needs huge data & compute

From Counting Words to Understanding Them

Bag of Words → TF-IDF → Embeddings → RNN / LSTM → Attention → Transformers → LLMs

Thank You

GEO TAG PHOTO :